

Here we have to develop a simple **railway ticket booking application**.

Develop a Railway Reservation Application contains

1. AC coach
2. Non AC coach
3. Seater

Each should contain 60 seats and 10 waiting list max allowed rest request should be cancelled.

You should have

1. Ticket Booking
2. Ticket Cancellation
3. Status Checking

Here they asked us to create a "Railway reservation system" and gave us 4 modules.
The modules were:

1. Booking
2. Availability checking
3. Cancellation
4. Prepare chart

We were asked to create the modules for representing each data first and to continue with the implementation phase.

A matrix game was given with 5 rules. We were asked to implement each of the rules separately.

```
R3 | -  -  -  |
R2 | -  -  -  |
R1 | -  -  -  |
    C1  C2  C3
```

Each of the 9 cells can either be empty or filled with an atom. R3, R2, R1 are the rays that originate from the left. C1, C2, C3 are the rays that originate from the bottom of the box.

Input : Position of the atoms and the rays that gets originated from the outside of the box.

```
Eg.) 3
      3 1
      2 2
      1 3
      3
      R3 C1 C3
```

Output : Print the box.

Rule 1:

A ray that has an atom in its path should print 'H' (Hit) If it does not have any atoms in its path, the ray should pass to the other side.

```
      C1      C3
R3 | -  -  -  | R3
H  | -  X  -  |
R1 | -  -  -  | R1
      C1  H  C3
```

Rule 2 & 3:

A ray that has an atom in its diagonal adjacent position should refract.

```

H | - - - |
H | X - - |
R | - X - |
   R H R

```

Input rays: R1, R2, C3

```

H | - X - |
R2 | - - - | C3
   | - - - |
   R2 C3

```

Rule 4:

A ray that has atoms in both of the diagonal adjacent positions should reflect back.

Input ray: C2

```

| - - - |
| X - X |
| - - - |
   R

```

Input ray: R2

```

| - X - |
R | - - - |
   | - X - |

```

Rule 5:

The deflection of rays should happen in the order of the input rays.

Input Rays: R3, R2, C1, C3

```

H | - X - |
R2 | - - - | C3
   | - - - |
   R2 C3

```

The final task was to implement these rules for dynamic matrix size.

Input : no of rows, no of columns

```

Eg.) 4 4 (row & column)
      2 (No of atoms)
      4 4 (Position of atom)
      2 2 (Position of atom)
      2 (No of rays)
      R4 C2 (Ray number)

```

```

H | - - - X |
  | - - - - |
  | - X - - |
  | - - - - |
    H

```

The final task was very confusing and it had to handle all the cases. There are chances for a ray to end at the starting position if the number of rows and columns are more than 5.